

SAM JONATHAN J

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SUMMARY

Computer Science graduate with a B.Tech in CSE and an ongoing M.Sc. in Data Science, with practical experience spanning data analysis, machine learning, AI engineering, and full stack development. Proficient in Python, SQL, R, and TypeScript, with proven ability to build end-to-end solutions from predictive models and NLP pipelines to computer vision systems and web applications. Certified in AI, data analytics, and Python by Google and IBM.

TECHNICAL SKILLS

- **Programming:** Python, SQL, R, HTML, TypeScript
- **AI/ML:** Machine Learning, Deep Learning, NLP, Computer Vision, YOLOv8, PyTorch, Hugging Face Transformers (BART, T5, Pegasus), Ollama, TinyLlama
- **Frameworks & Tools:** Streamlit, React (Vite), OpenCV, pyttsx3, SpeechRecognition, gTTS, Pydub, WeatherAPI
- **Data & Analytics:** Tableau, Excel, Predictive Modeling, Structured & Unstructured Data Processing
- **Strengths:** Problem Solving, Logical Reasoning, Self Directed Learning, Time Management

EDUCATION

Symbiosis International (Deemed University), Pune

M.Sc. Data Science

Sep 2025 – Sep 2027 (Pursuing)

Karunya Institute of Technology and Sciences, Coimbatore

B.Tech, Computer Science and Engineering CGPA: 7.55/10

Sep 2021 – Apr 2025

EXPERIENCE

CISCO

Remote

Python Programming Intern

3-Month Program

- Completed PCAP – Programming Essentials in Python, covering Python 3, OOP, data structures, control flow, and functions, building the programming foundation required for AI and ML engineering work.
- Applied debugging, problem solving, and logical thinking across practical coding exercises, reinforcing the depth needed to build and maintain production AI systems.

PROJECTS

Pathfinder: AI-Powered Obstacle Detection & Navigation System – [GitHub](#)

Python | YOLOv8 | PyTorch | OpenCV | Streamlit | gTTS | WeatherAPI

- Built a real-time obstacle detection system for the visually impaired using YOLOv8 and PyTorch, detecting **80+ object classes** at **24+ fps** on a standard laptop CPU.
- Integrated asynchronous voice alerts (gTTS / Pydub) with a 5-second smart cooldown, cutting redundant audio notifications by **~80%** without disrupting the live video pipeline.
- Delivered a Streamlit dashboard with configurable confidence thresholds and WeatherAPI integration, reducing user setup time to under **30 seconds** from launch to active detection.

Local Offline Voice Chatbot – [GitHub](#)

Python | Ollama (TinyLlama) | Streamlit | pytsx3 | SpeechRecognition

- Built a **100% offline** AI chatbot using Ollama and TinyLlama, achieving average response latency under **3 seconds** on a standard laptop with zero cloud dependency.
- Implemented bidirectional voice interaction supporting **full multi-turn context**, combining Speech Recognition input and pytsx3 TTS output with session state history management.
- Delivered a single-click Streamlit UI with custom CSS, eliminating all post-install configuration and reducing time-to-first-response to under **10 seconds**.

YouTube Transcript Summarizer

Python | Hugging Face Transformers (BART, T5, Pegasus) | YouTube Transcript API

- Built an NLP pipeline capable of summarising videos up to **60+ minutes**, compressing transcripts to summaries **85–90% shorter** than the original text.
- Benchmarked **3 Transformer architectures** (BART, T5, Pegasus) side-by-side, enabling users to select the most coherent output for lectures, interviews, and tutorials.

Workout Tracker Web Application – [GitHub](#)

React (Vite) | TypeScript | Custom CSS

- Built a **fully responsive** component-based frontend supporting logging of sets, reps, and weights across **multiple workout types**, with React hooks managing all dynamic state.
- Designed **cross-device compatible** UI with modular, reusable components, ensuring consistent experience across mobile and desktop without any additional frameworks.

CERTIFICATIONS

- Google – Professional Certificate: Google Data Analytics
- IBM – AI Developer Professional Certificate
- Google – Introduction to Artificial Intelligence
- Google – Python for Data Science, AI and Development
- OpenEDG – Python Programming Essentials (PCAP)